

Fourier Series

$$f(x) = \sum_{k=0}^{\infty} \alpha_k \cos(\overbrace{\frac{2\pi k}{L} \cdot x}^{\omega \cdot x})$$

from $x=0$ to $x=L$
for even function of f .

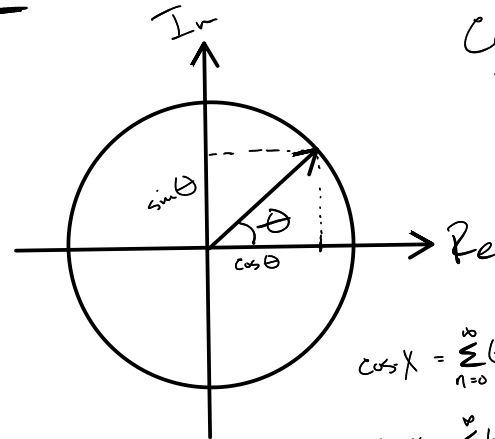
$$f(x) = \sum_{k=1}^{\infty} \beta_k \sin\left(\frac{2\pi k}{L} \cdot x\right)$$

from $x=0$ to $x=L$
for odd functions of f .

$$f(x) = \sum_{k=0}^{\infty} \alpha_k \cos\left(\frac{2\pi k}{L} \cdot x\right) + \sum_{k=1}^{\infty} \beta_k \sin\left(\frac{2\pi k}{L} \cdot x\right) \quad \leftarrow \text{for any function from } 0 \text{ to } L$$

$$e^{i \frac{2\pi k}{L} x} = \cos\left(\frac{2\pi k}{L} x\right) + i \cdot \sin\left(\frac{2\pi k}{L} x\right)$$

$$\underline{e^{i\theta}} = \cos\theta + i \sin\theta$$



Complex Plane

Real Numbers
2, -18, 9.8, $\sqrt{2}$

Imaginary Numbers
 $i, -i, \sqrt{2}i,$

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$$

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

$$e^{ix} = \sum_{n=0}^{\infty} \frac{ix^n}{n!}$$

$$f(x) = \sum_{k=-\infty}^{+\infty} \gamma_k e^{i \left(\frac{2\pi k}{L} x \right)}$$

